
ME 613: Thermal Radiation

Course Code	ME 613
Course Name	Thermal Radiation
L-T-P-C	3-0-0-3
Prerequisites	Fundamentals of heat transfer
Intended for	MS/PhD
Distribution	Elective
One-time approval	recommended by the 9th BoA (Oct 2015)

Course Contents

- **Fundamentals of Thermal Radiative Heat Transfer**

Introduction, The nature of thermal radiation, Basic laws of thermal radiation Introduction to radiation characteristics of opaque surfaces, gases, solid, liquid and particles. [3 Lectures]

- **Radiation properties of real surfaces**

Definitions of radiation phenomena at surfaces, Radiative properties of metals, non-conductors, semi-transparent sheets [2 Lectures]

- **Radiation Exchange between surfaces**

Introduction to view factors and methods to calculate the view factors. Radiative exchange between Grey, Diffuse Surfaces, Partially-Specular Grey Surfaces. [10 Lectures]

- **Gas Radiation**

The equation of radiative transfer in participative media, Boundary conditions for the equation of transfer, divergence of radiative heat flux, overall energy conservation, radiative properties of molecular gases, line radiation, spectral models for radiative transfer calculations, Narrow band models, Wide band models. radiative properties of particulate media Rayleigh scattering. P1, Zonal and Finite volume method for radiative transfer equation. [20 Lectures]

- **Radiation properties of semi-transparent media and Collimated beam radiation**

Radiation phenomena in semi-transparent solids and liquids. Monte carlo method for the collimated beam radiation. [4 Lectures]

- **Design and Performance study of solar conversion devices by using open source or standard soft-wares [3 Lectures]**

Text and Reference Books

1. M. F. Modest, **Radiative Heat Transfer**, 3rd Edition, Academic Press, 2013.
2. J. R. Howell, R. Siegel, M.P. Menguc, **Thermal Radiation Heat Transfer** 5th Edition, CRC Press, 2011.